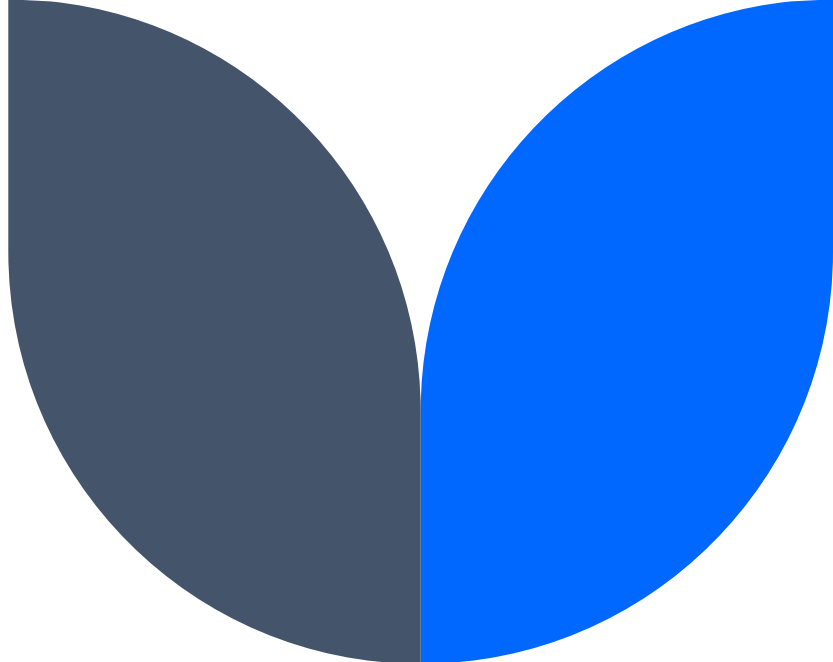
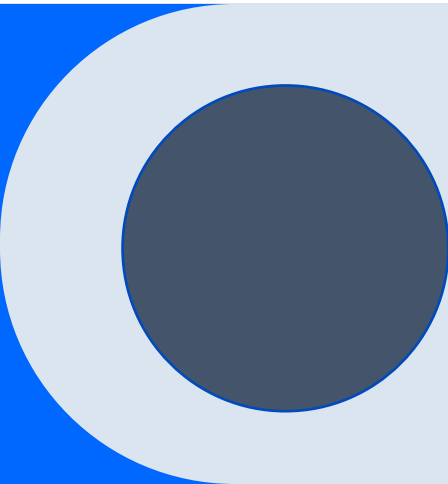




Insurance Analysis

Karen Podier



Dataset Overview

Business Context:

The purpose of this analysis is to explore the customers' demographic and health information along with their medical insurance charges. The aim is to identify which factors influence insurance charges the most. These insights will help in pricing policies and advising customers.

The dataset:

The dataset consists of 1338 rows and 7 columns. These columns include:

- Age - Age of the insured person
- Sex - Gender of the insured person (male, female)
- BMI - Body Mass Index (an indicator of body fat based on height and weight)
- Children - Number of children/dependents covered by the insurance
- Smoker Status - Whether the person is a smoker (yes, no)
- Region - Residential region of the insured (northwest, southeast, etc.)
- Charges - Medical insurance cost billed to the individual



Data Preparation

This project was completed with python, and I used pandas to inspect the dataset. I reviewed the summary statistics and confirmed the data types were appropriate for the columns and further analysis.

I checked for missing values and there were none found. I identified 1 duplicate row and removed it using `drop_duplicates()`. Once I checked the duplicate was removed, the dataset was cleaned and ready for analysis.

Two columns were added for analysis:

- **BMI Category** – Customers group as underweight, normal, overweight or obese
- **Age Group** – Customers grouped as young, middle-aged or senior

```
[11]: import pandas as pd

df = pd.read_csv(r"C:\Users\kpodi\Downloads\insurance.csv")

df.head()
df.info()
df.describe()
df.shape
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1338 entries, 0 to 1337
Data columns (total 7 columns):
 #   Column      Non-Null Count  Dtype
---  -
 0   age         1338 non-null   int64
 1   sex         1338 non-null   object
 2   bmi         1338 non-null   float64
 3   children    1338 non-null   int64
 4   smoker      1338 non-null   object
 5   region      1338 non-null   object
 6   charges     1338 non-null   float64
dtypes: float64(2), int64(2), object(3)
memory usage: 73.3+ KB
```

```
[11]: (1338, 7)
```

```
[12]: df.duplicated().sum()
```

```
[12]: np.int64(1)
```

```
[13]: df = df.drop_duplicates()
```

```
[14]: df.duplicated().sum()
```

```
[14]: np.int64(0)
```

Overview of Charges

Average charge:
\$13,279

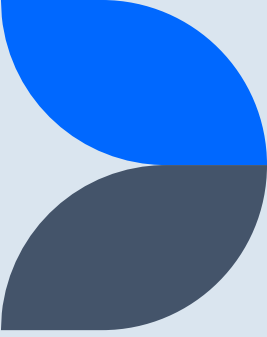
There is significant variation in charges billed to customers, indicating that certain factors may strongly impact insurance costs.

Minimum charge:
\$1,122

These possible factors will be explored throughout this project.

Maximum charge:
\$63,770

Customers Profile: Smoking Status

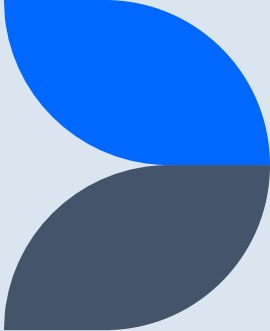


Smoker Status	Count
No	1063
Yes	274

The dataset contains more non-smokers than smokers.

This gives useful context before comparing insurance charges by smoking status, as group size can affect how patterns are interpreted.

Average Charges by Customer Groups



Charges by Sex:

Sex	Average Charge (\$)
Female	12,570
Male	13,975

Charges by Region:

Region	Average Charge (\$)
North-East	13,406
North-West	12,451
South-East	14,735
South-West	12,347

Charges by Number of Children:

Number of Children	Average Charge (\$)
0	12,385
1	12,731
2	15,074
3	15,355
4	13,851
5	8,786

Average charges were compared across sex, region and number of children to identify early patterns in the data.

The differences by sex and region were present, but not very large. The number of children showed some variation, although the pattern was not fully consistent.

Data Transformation

To support deeper analysis, two new categorical columns were created from existing data.

BMI Category: Customers were grouped as Underweight, Normal, Overweight or Obese based on their BMI values.

Age Group: Customers were grouped as Young, Middle-Aged or Senior based on their age.

```
df[['bmi', 'bmi_category']].head()
```

```
[27]:
```

	bmi	bmi_category
0	27.900	Overweight
1	33.770	Obese
2	33.000	Obese
3	22.705	Normal
4	28.880	Overweight

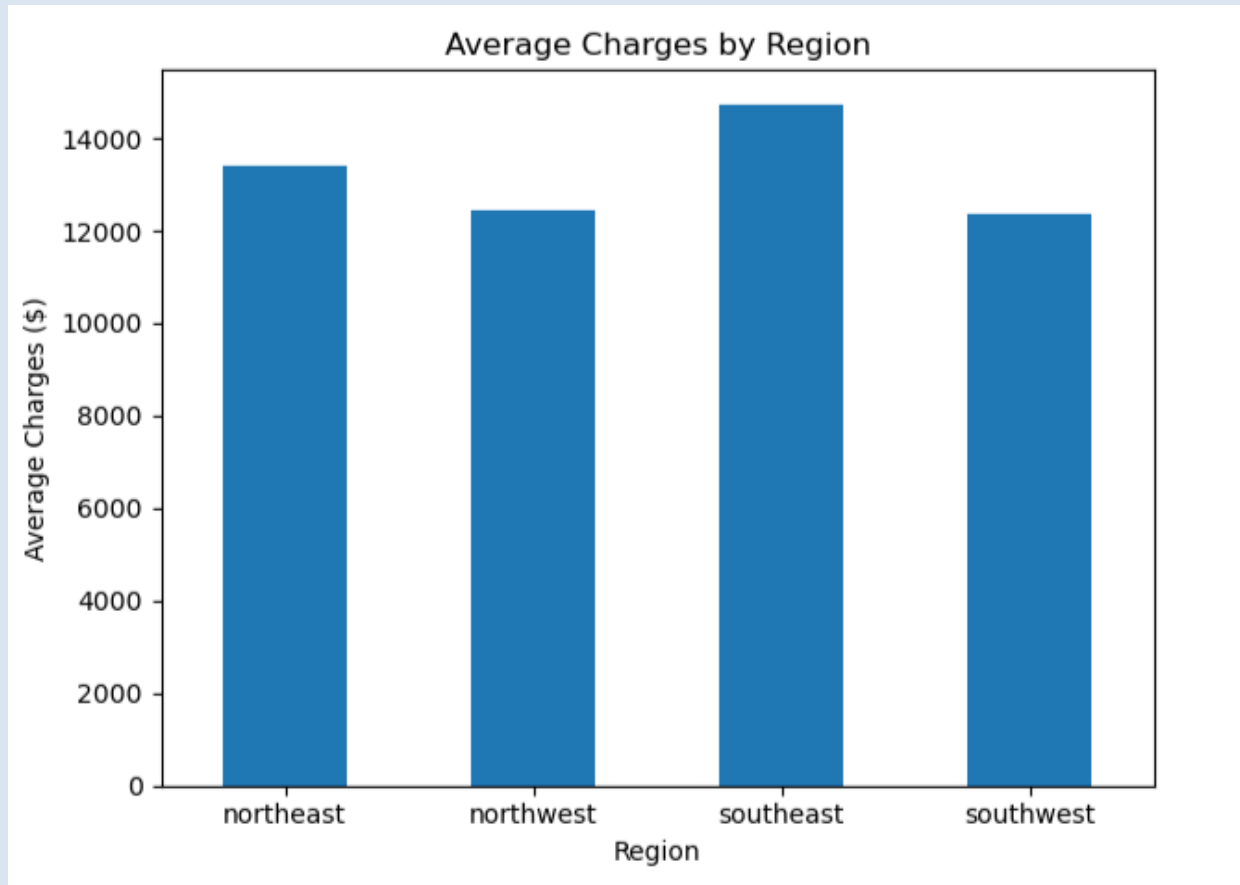
```
df[['age', 'age_group']].head()
```

```
[28]:
```

	age	age_group
0	19	Young
1	18	Young
2	28	Young
3	33	Middle-Aged
4	32	Middle-Aged

Average Charges by Region

This bar chart compares the average insurance charges across the four regions.



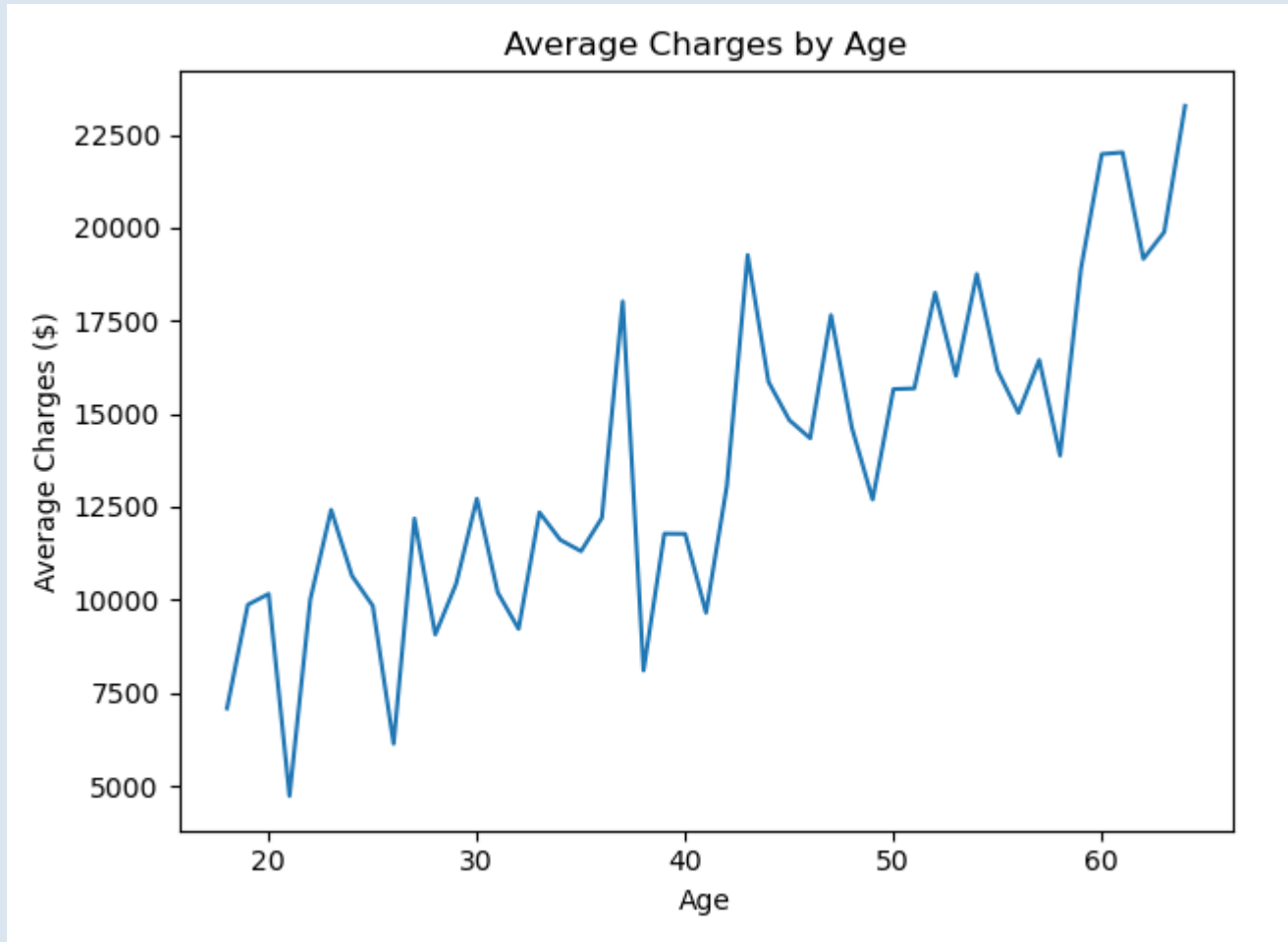
The southeast region had the highest average charge at approximately \$14,735, while the southwest had the lowest at approximately \$12,347.

However, the regional differences are relatively small compared with the overall range of charges, suggesting that region may not be the strongest driver of insurance costs.

Insight: Region affects charges slightly, but other factors are likely to have a stronger impact.

Average Charges by Age

This line chart shows how average insurance charges change with age.

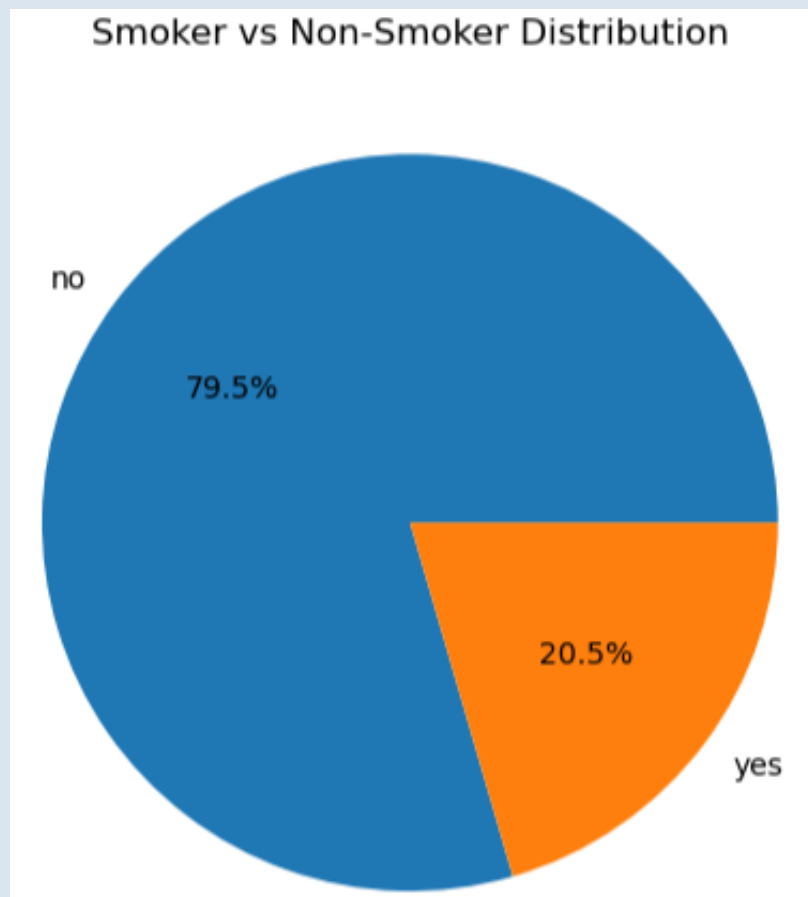


Overall, charges tend to increase as age increases. Although there are some fluctuations between individual ages, the general trend suggests that older customers are billed higher insurance charges.

Insight: Age appears to be an important factor influencing insurance charges.

Smoker vs Non-Smoker Distribution

This pie chart shows the proportion of smokers and non-smokers in the dataset.

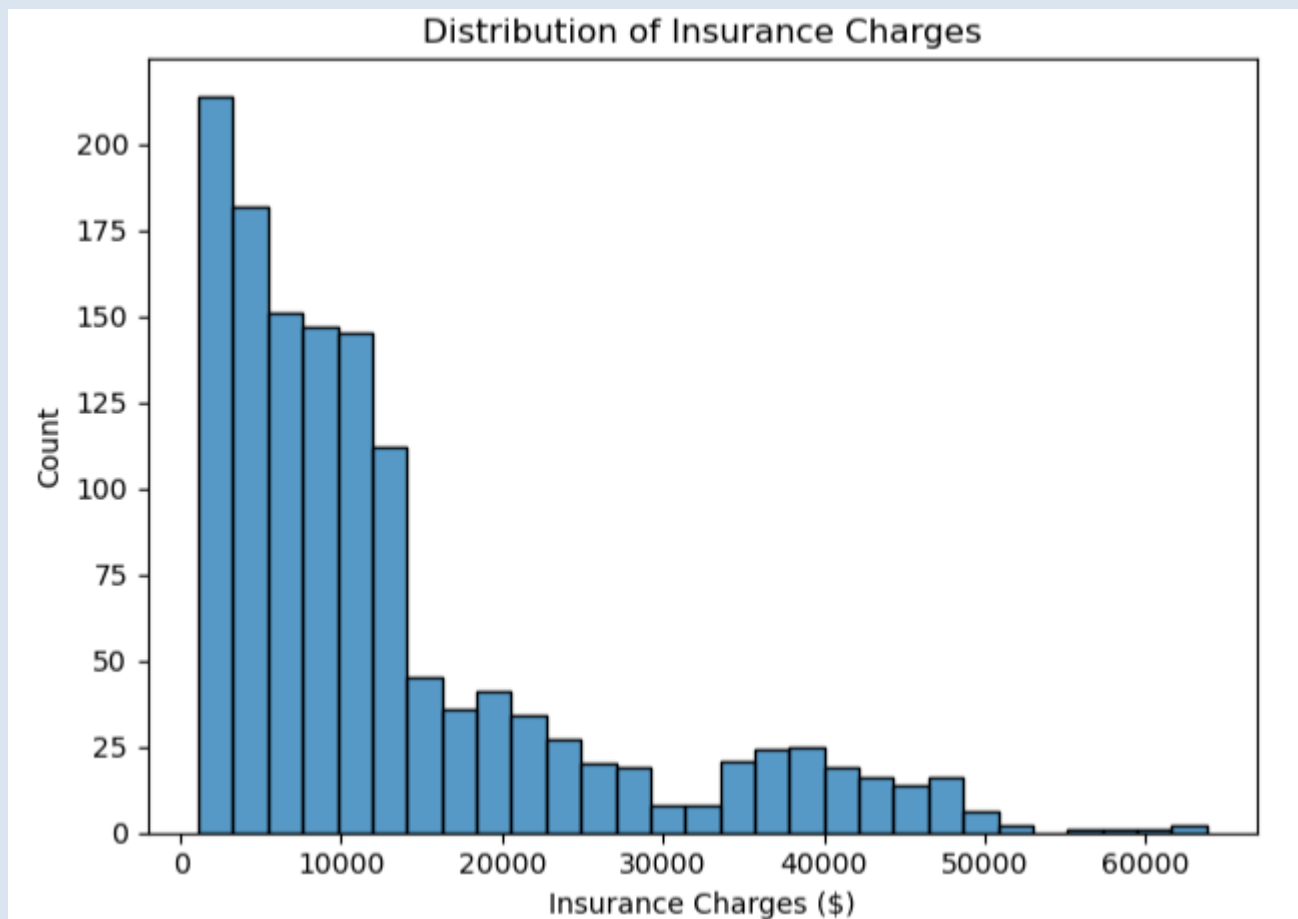


Most customers were non-smokers, making up 79.5% of the dataset, while smokers made up 20.5%.

Insight: Although smokers make up a smaller proportion of the dataset, smoking status may still be an important factor influencing insurance charges.

Distribution of Insurance Charges

This histogram shows how insurance charges are distributed across the dataset.

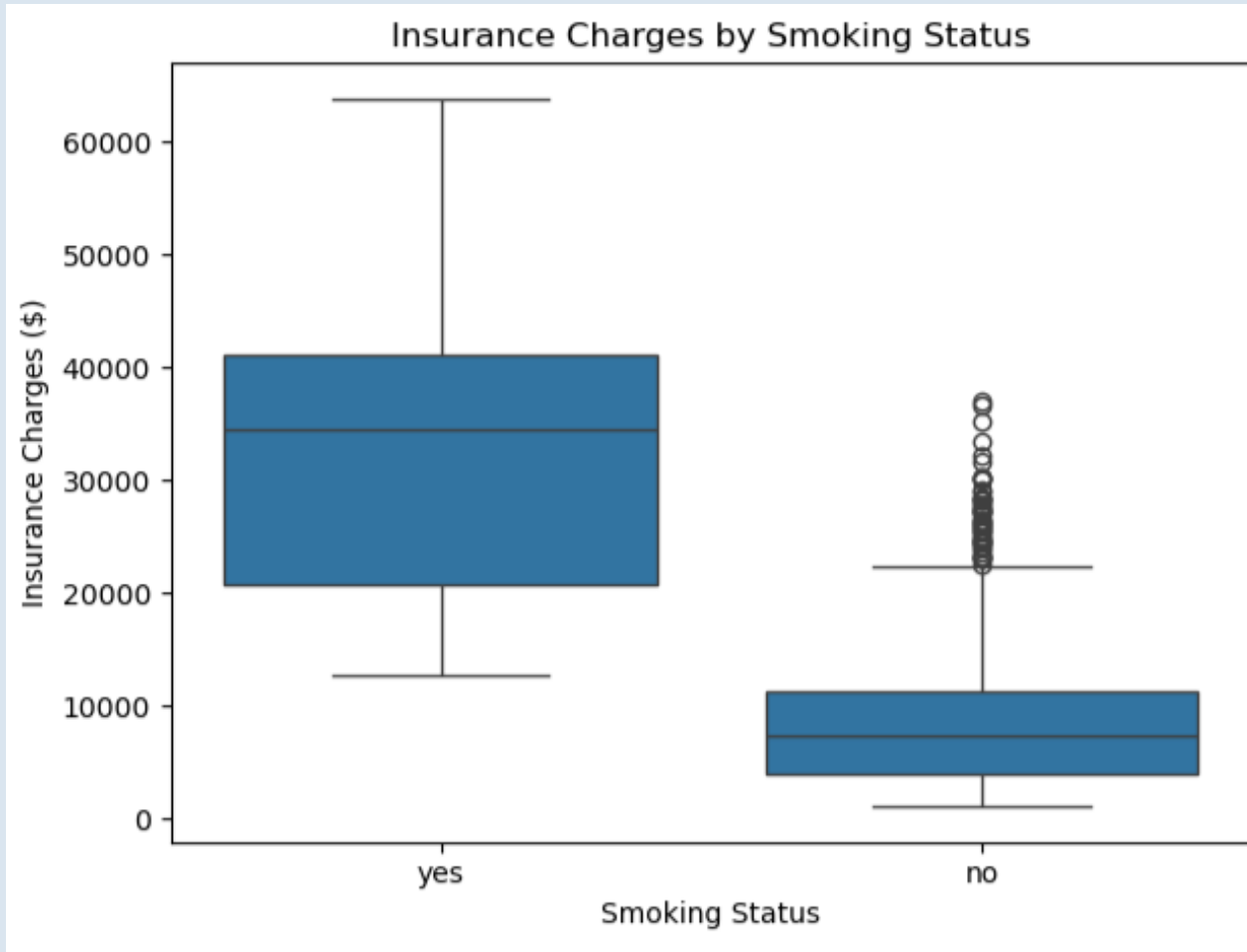


Most customers have lower insurance charges, with many values concentrated below approximately \$15,000. A smaller number of customers have much higher charges, creating a right-skewed distribution.

Insight: The dataset contains a small group of customers with very high charges, which suggests that certain risk factors may be linked to much higher insurance costs.

Insurance Charges by Smoking Status

This boxplot compares the distribution of insurance charges for smokers and non-smokers.



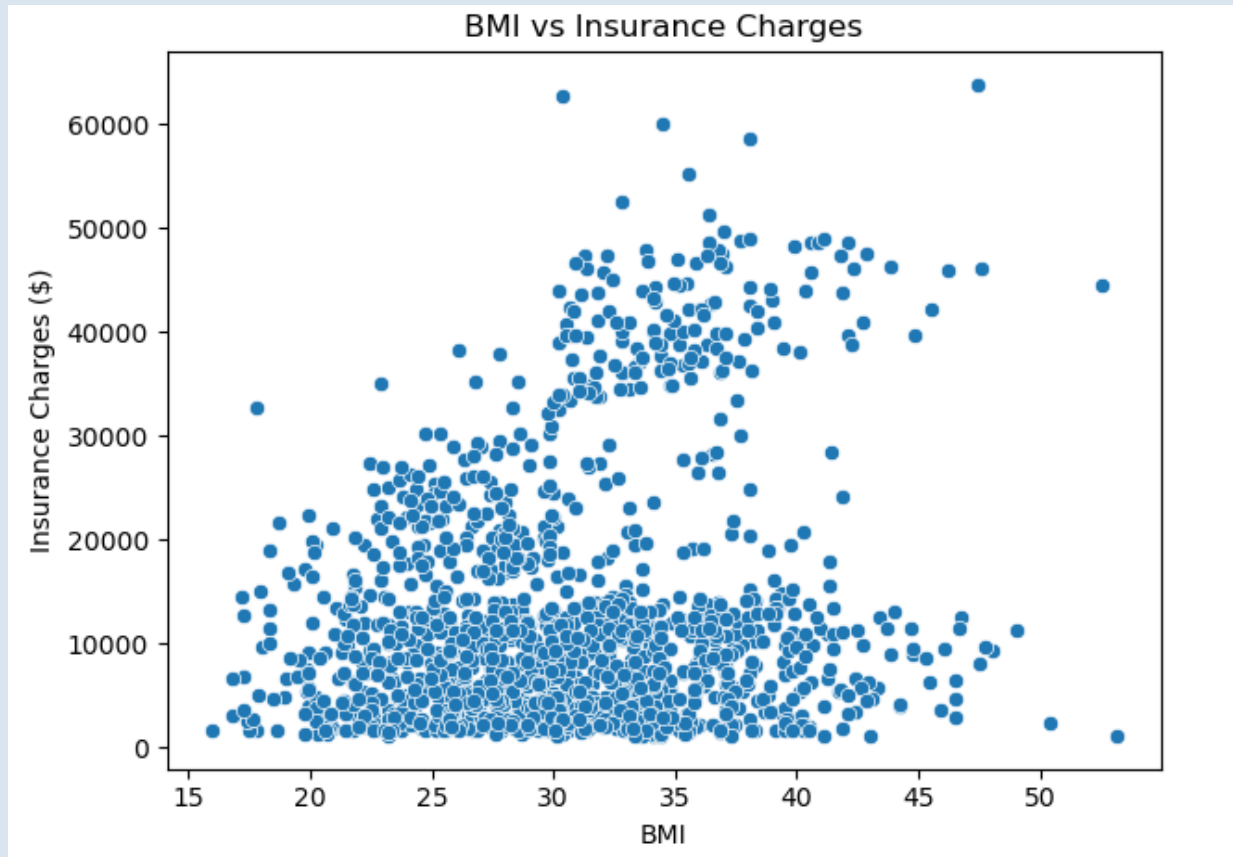
Smokers have noticeably higher insurance charges than non-smokers. The median charge for smokers is much higher, and the range of charges is wider compared with non-smokers.

Although smokers make up a smaller proportion of the dataset, they are linked with significantly higher insurance costs.

Insight: Smoking status appears to be one of the strongest factors influencing insurance charges.

BMI vs Insurance Charges

This scatter plot compares customers' BMI values with their insurance charges.



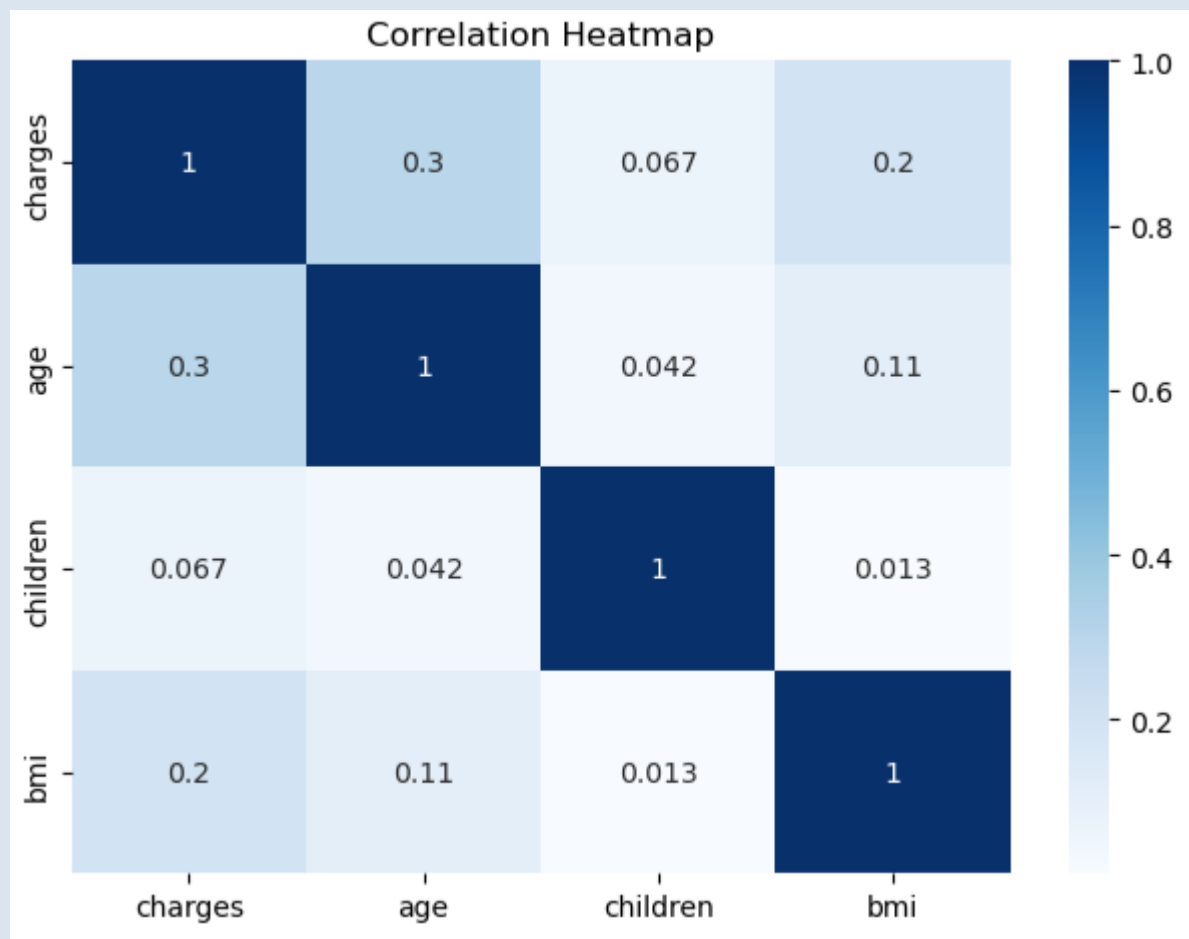
The chart shows that BMI alone does not have a clear linear relationship with insurance charges. Many customers have lower charges across different BMI levels.

However, the highest charges appear more often among customers with higher BMI values, particularly around BMI 30 and above.

Insight: BMI may influence insurance charges, but its impact is likely stronger when combined with other factors such as smoking status or age.

Correlation Heatmap

This heatmap shows the relationships between the numerical variables in the dataset.



Age had the strongest positive correlation with insurance charges at 0.30, followed by BMI at 0.20.

The number of children had a very weak correlation with charges at 0.067, suggesting it has less influence on insurance costs compared with age and BMI..

Insight: Among the numerical variables, age appears to have the strongest relationship with insurance charges.

Key Insights

- Smoking status showed the clearest difference in insurance charges. Smokers had noticeably higher charges than non-smokers.
- Age had the strongest correlation with charges among the numerical variables, suggesting charges generally increase as customers get older.
- BMI may influence charges, especially at higher BMI values, but the relationship is not as clear on its own.
- Region and number of children showed some variation, but they appeared to have a weaker influence compared with smoking status, age and BMI.

Overall finding: Smoking status, age and BMI appear to be the most important factors influencing insurance charges.

Business Recommendations



Smoking Status:

Use smoking status as a key pricing risk factor.

Smoking status showed the clearest difference in insurance charges, so it should remain an important part of risk assessment and pricing decisions.



Age:

Consider age-based risk trends when reviewing policies.

Charges generally increased with age, so age should be considered when estimating likely healthcare costs.



BMI and Wellness:

Offer wellness support for customers with higher BMI.

BMI may contribute to higher charges, especially when combined with other factors. Offering health and wellness support could help reduce future risk.

Closing and Reflections

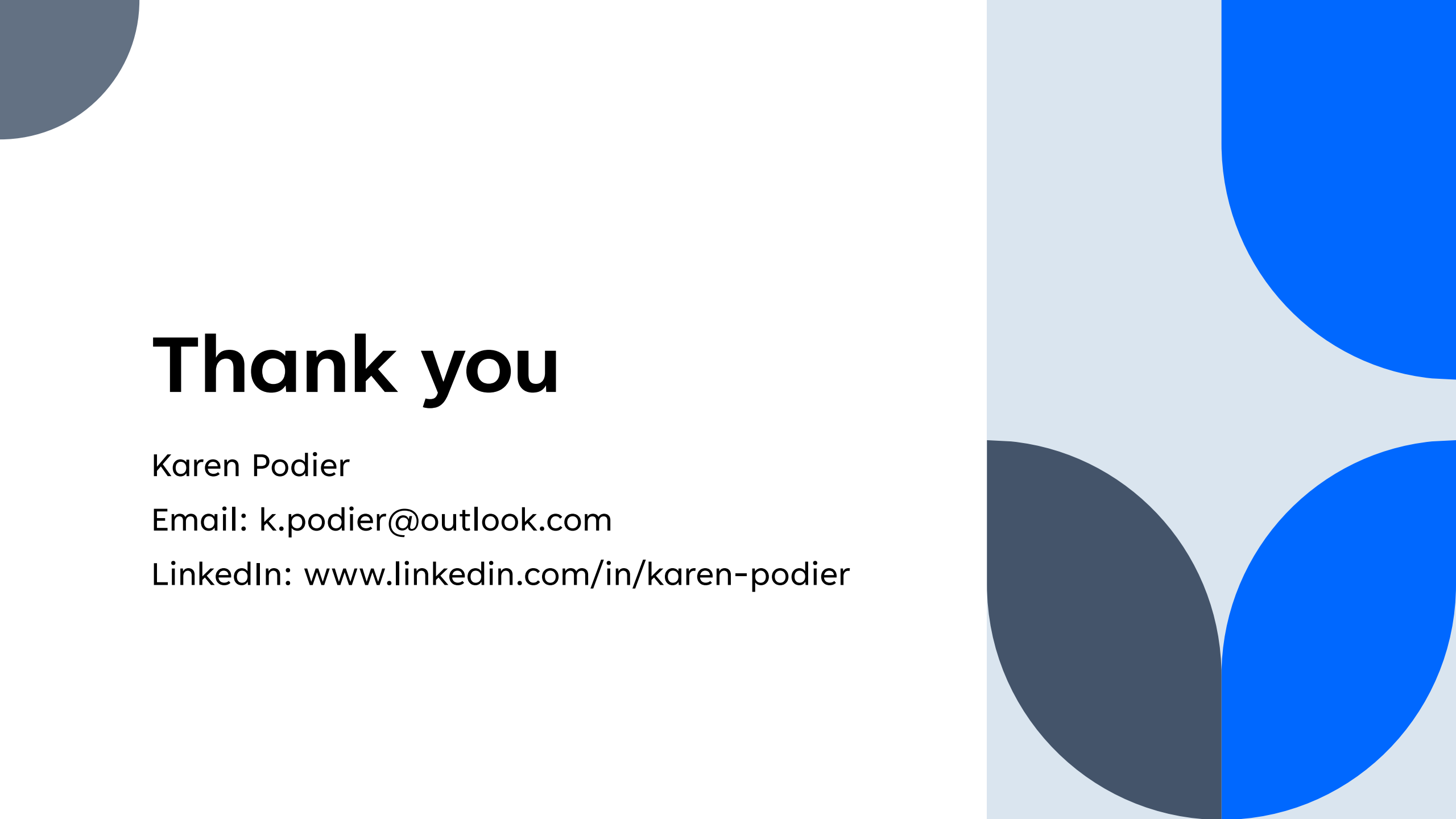
This project explored how customer demographic and health factors relate to medical insurance charges.

The analysis included data cleaning, exploratory analysis, data transformation, and visualisations using Matplotlib and Seaborn.

Reflection:

The three strongest drivers of insurance charges appear to be smoking status, age, and BMI. Smoking status showed the clearest difference in charges, age had the strongest numerical correlation with charges, and BMI appeared to be linked with higher charges especially when combined with other risk factors.





Thank you

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